第二章 线性模型

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本章内容

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  - 基本形式
  - Linear Regression with One Variable
    - 解析解
    - 梯度下降法
  - Linear Regression with Multiple Variables
    - 解析解
    - 梯度下降法
- Logistic Regression:Classification
  - 二分类
  - 多分类
- 性能度量

线性回归——基本形式

单变量（单特征或单属性）：

Size (feet²)	Price ($1000)
x	y
2104	460
1416	232
1534	315
852	178
…	…

多变量（多特征或多属性）：

Size (feet²) x_1	Number of bedrooms x_2	Number of floors x_3	Age of home (years) x_4	Price ($1000) y
2104	5	1	45	460
1416	3	2	40	232
1534	3	2	30	315
852	2	1	36	178
…	…	…	…	…

$$y_i \approx f(x_i) = w\,x_i + b$$

$$y_i \approx f(\boldsymbol{x}_i) = w_1 x_{i1} + w_2 x_{i2} + w_3 x_{i3} + w_4 x_{i4} + b$$

一般情况下：$y_i \approx f(\boldsymbol{x}_i) = w_1 x_{i1} + w_2 x_{i2} + \cdots + w_n x_{in} + b = \boldsymbol{w}^T \boldsymbol{x}_i + b$

其中：$\boldsymbol{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}$ $\boldsymbol{x}_i = \begin{bmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{in} \end{bmatrix}$

线性回归——基本形式

或记作：

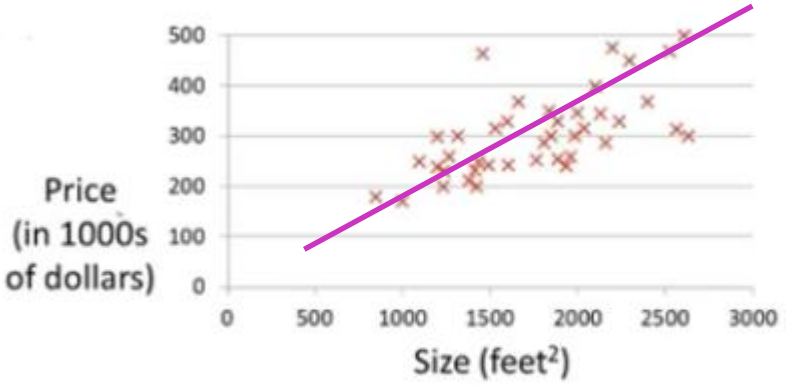
$$\begin{aligned} y_i &\approx f(\boldsymbol{x}_i) = w_1 x_{i1} + w_2 x_{i2} + \cdots + w_n x_{in} + b \\ &= w_0 x_{i0} + w_1 x_{i1} + \cdots + w_n x_{in} \qquad (w_0 = b, x_{i0} = 1) \\ &= \mathbf{w}^{\boldsymbol{T}} \mathbf{x}_i \end{aligned}$$

其中：

$$\mathbf{w} = \begin{bmatrix} b \\ w_1 \\ \vdots \\ w_n \end{bmatrix} \qquad \mathbf{x}_i = \begin{bmatrix} 1 \\ x_{i1} \\ \vdots \\ x_{in} \end{bmatrix}$$

单变量线性回归

Size (feet²)	Price ($1000)
x	y
2104	460
1416	232
1534	315
852	178
…	…



m： #训练样本

x ： 输入变量或特征

y ： 输出变量或目标变量

(x, y)： 单个训练样本

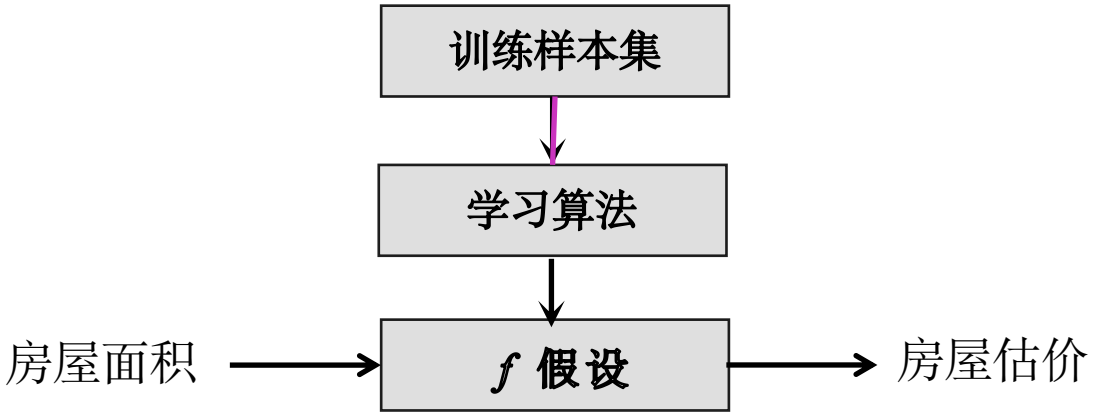
(x_i, y_i)： 第i个训练样本

例：

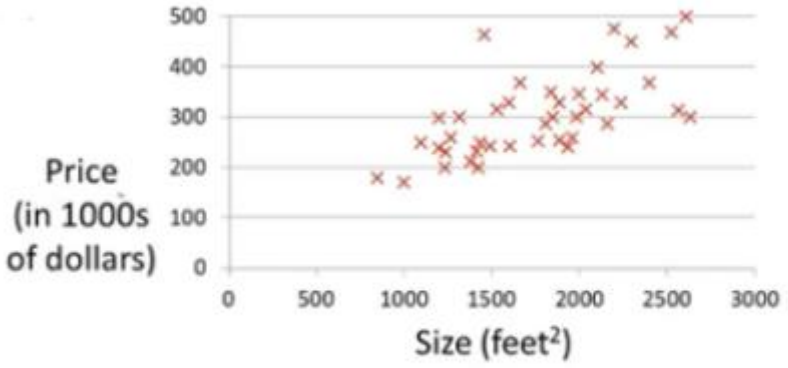
$x_1 = 2104$ $y_1 = 460$

$x_2 = 1416$ $y_2 = 232$

单变量线性回归



f如何得到？ $f(x_i) = w\,x_i + b \cong y_i$



单变量线性回归

训练集

Size in feet² (x)	Price ($) in 1000's (y)
2104	460
1416	232
1534	315
852	178
…	…

假设 $f(x_i) = w\,x_i + b$

参数 w, b

问题：如何选择 w, b ？

思想：对于训练集中的训练样本(x_i，y_i)，使得$f(x_i)$尽量接近y_i。

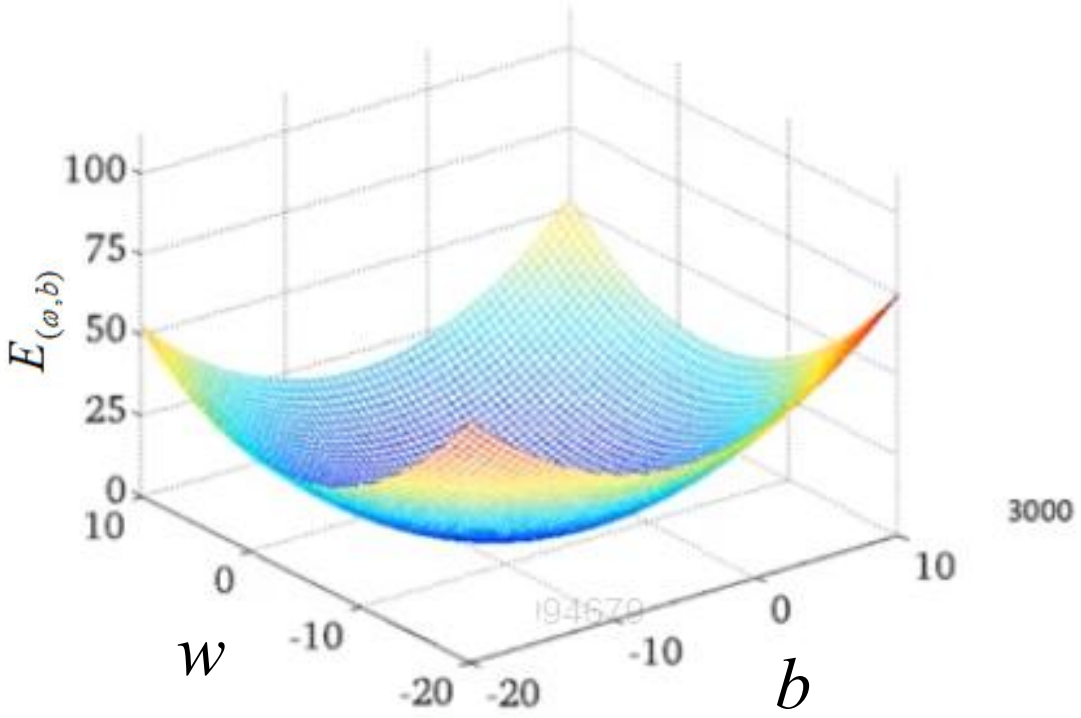
$$\min_{w,b} \frac{1}{2m}\sum_{i=1}^{m}(f(x_i)-y_i)^2 = \min_{w,b} \frac{1}{2m}\sum_{i=1}^{m}(y_i-f(x_i))^2$$

$$E_{(w,b)} = \frac{1}{2m}\sum_{i=1}^{m}(f(x_i)-y_i)^2$$ 损失函数或代价函数

目标： $\min_{w,b} E_{(w,b)}$

单变量线性回归

$E_{(w,b)}$是关于w和b的凸函数



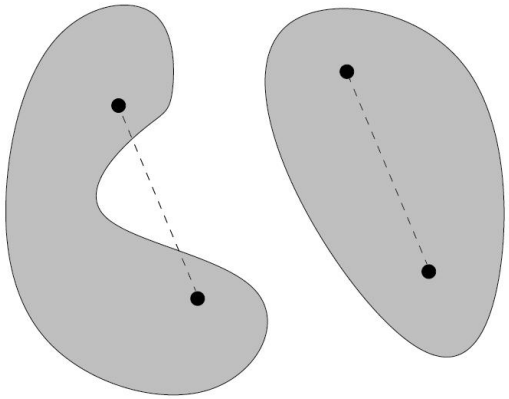
单变量线性回归

凸域的概念：若对区域D中任意两点α和β，　$\alpha \in D, \beta \in D$均有：

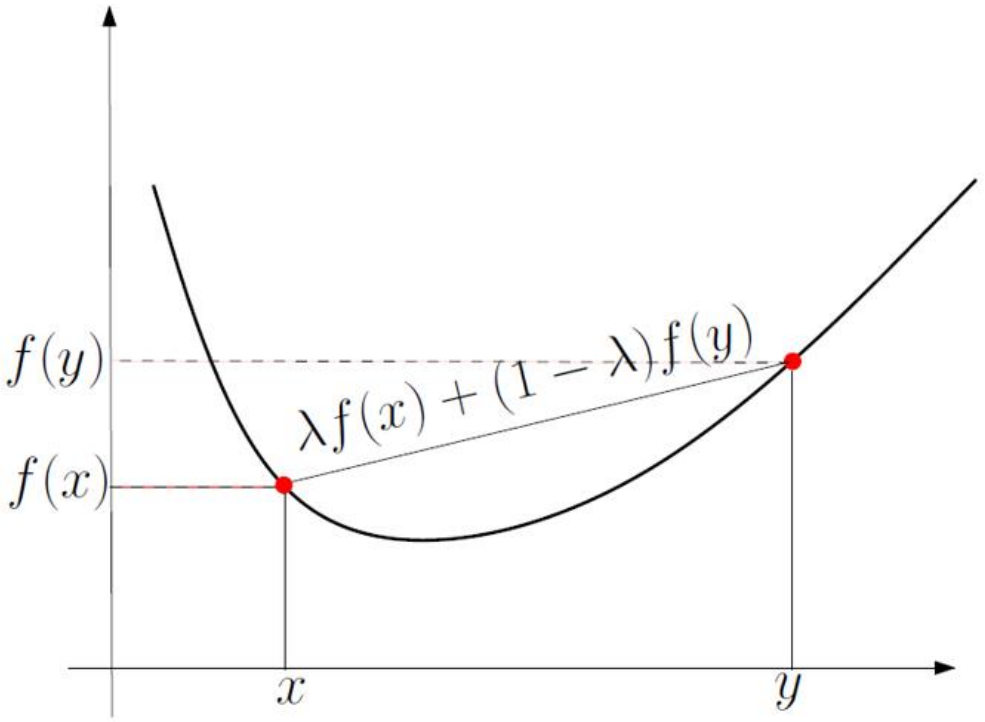
$$\lambda\alpha + (1-\lambda)\beta \in D, \forall 0 \le \lambda \le 1$$

则称：区域D是凸域。

理解：若两点α和β在凸域D内，则α和β之间的线段也整个在区域D内。



单变量线性回归



$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda) f(y)$$

单变量线性回归

一元函数判断凹凸性：

对区间D上定义的函数f，对于 $x_1 \in D, x_2 \in D, \theta \in [0,1]$

（1）$f(\theta x_1 + (1-\theta)x_2) \le \theta f(x_1) + (1-\theta) f(x_2)$，则称$f$为区间D上的凸函数；

（2）$f(\theta x_1 + (1-\theta)x_2) \ge \theta f(x_1) + (1-\theta) f(x_2)$，则称$f$为区间D上的凹函数.

一元凹凸函数求最值：

设$f(x)$是在开区域D内连续可微的凸（凹）函数，$x_0 \in D$，且$f'(x_0) = 0$，则$f(x_0)$必为$f(x)$在D内最小（大）值。

单变量线性回归

二元函数判断凹凸性：（尝试证明！）

设$f(x,y)$在区间D上具有二阶连续偏导数，记A=$f''_{xx}(x,y)$，B=$f_{xy}''(x,y)$，C=$f''_{yy}(x,y)$，则

（1）在D上恒有A>0，且$AC-B^2\geq 0$时，$f(x,y)$在D上是凸函数；

（2）在D上恒有A<0，且$AC-B^2\geq 0$时，$f(x,y)$在D上是凹函数。

二元凹凸函数求最值：

设$f(x,y)$是在开区域D内具有连续偏导数的凸（凹）函数，$(x_0,y_0)\in D$ 且$f'_x(x_0,y_0)=0$，$f'_y(x_0,y_0)=0$，则$f(x_0,y_0)$必为$f(x,y)$在D内最小（大）值。

$$E_{(w,b)} = \frac{1}{2m}\sum_{i=1}^{m}(y_i - f(x_i))^2 = \frac{1}{2m}\sum_{i=1}^{m}(y_i - wx_i - b)^2$$

$$E_{(w,b)} = \sum_{i=1}^{m}(y_i - f(x_i))^2 = \sum_{i=1}^{m}(y_i - wx_i - b)^2$$

$$\min_{w,b} E_{(w,b)}$$

方法一：求解析解

$$\frac{\partial E_{(w,b)}}{\partial w} = \frac{\partial \sum_{i=1}^{m}(y_i - wx_i - b)^2}{\partial w} = \sum_{i=1}^{m}\frac{\partial (y_i - wx_i - b)^2}{\partial w} = 2\sum_{i=1}^{m}(y_i - wx_i - b)(-x_i)$$

$$= -2\sum_{i=1}^{m}\left((y_i - b)x_i - wx_i^2\right) = 2\left(w\sum_{i=1}^{m}x_i^2 - \sum_{i=1}^{m}(y_i - b)x_i\right) \cdots \quad (3.5)$$

$$\frac{\partial E_{(w,b)}}{\partial b} = \sum_{i=1}^{m}2(y_i - wx_i - b)(-1) = 2\left(\sum_{i=1}^{m}b - \sum_{i=1}^{m}(y_i - wx_i)\right)$$

$$= 2\left(mb - \sum_{i=1}^{m}(y_i - wx_i)\right) \cdots \quad (3.6)$$

$$\frac{\partial E_{(w,b)}}{\partial b} = 2\left(mb - \sum_{i=1}^{m} (y_i - wx_i) \right) = 0 \quad \Rightarrow b = \frac{1}{m} \sum_{i=1}^{m} (y_i - wx_i) \ \cdots \qquad (3.8)$$

$$\frac{\partial E_{(w,b)}}{\partial w} = 2\left(w \sum_{i=1}^{m} x_i^2 - \sum_{i=1}^{m} (y_i - b) x_i \right) = 0$$

$$\Rightarrow \quad w \sum_{i=1}^{m} x_i^2 = \sum_{i=1}^{m} (y_i - b) x_i = \sum_{i=1}^{m} y_i x_i - b \sum_{i=1}^{m} x_i = \sum_{i=1}^{m} y_i x_i - bm \frac{1}{m} \sum_{i=1}^{m} x_i$$

$= \bar{x}$

$$= \sum_{i=1}^{m} y_i x_i - bm\bar{x} = \sum_{i=1}^{m} y_i x_i - m\bar{x} \frac{1}{m} \sum_{i=1}^{m} (y_i - wx_i)$$

$= b$

$$= \sum_{i=1}^{m} y_i x_i - \bar{x} \sum_{i=1}^{m} y_i + \bar{x} w \sum_{i=1}^{m} x_i \ = \sum_{i=1}^{m} y_i (x_i - \bar{x}) + \frac{1}{m} \sum_{i=1}^{m} x_i \, w \sum_{i=1}^{m} x_i$$

$= \bar{x}$

$$= \sum_{i=1}^{m} y_i (x_i - \bar{x}) + \frac{w}{m} \left(\sum_{i=1}^{m} x_i \right)^2$$

$$\Rightarrow \qquad w = \frac{\sum_{i=1}^{m} y_i (x_i - \bar{x})}{\sum_{i=1}^{m} x_i^2 - \frac{1}{m} \left(\sum_{i=1}^{m} x_i \right)^2} \qquad \cdots \qquad (3.7)$$

$$E_{(w,b)} = \frac{1}{2m}\sum_{i=1}^{m}(f(x_i)-y_i)^2$$

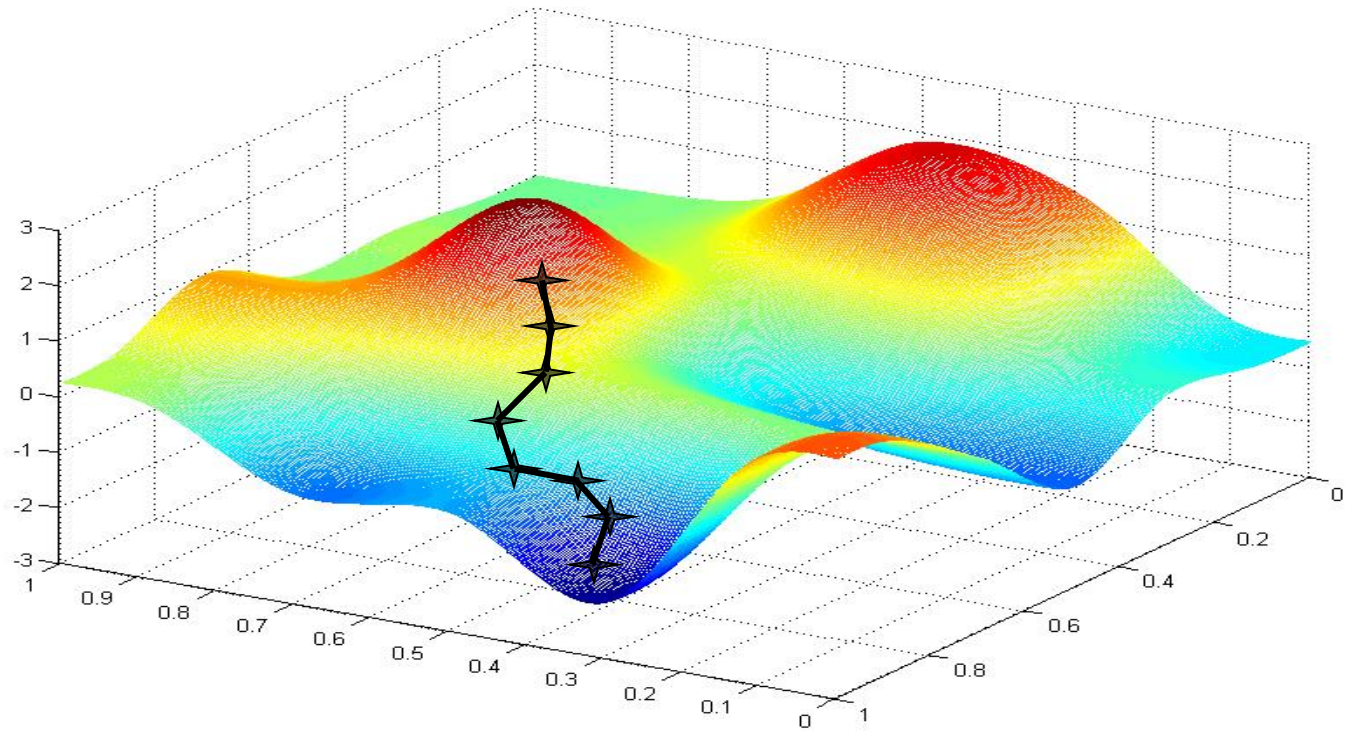
$$\min_{\omega,b} E_{(\omega,b)}$$

方法二：梯度下降法

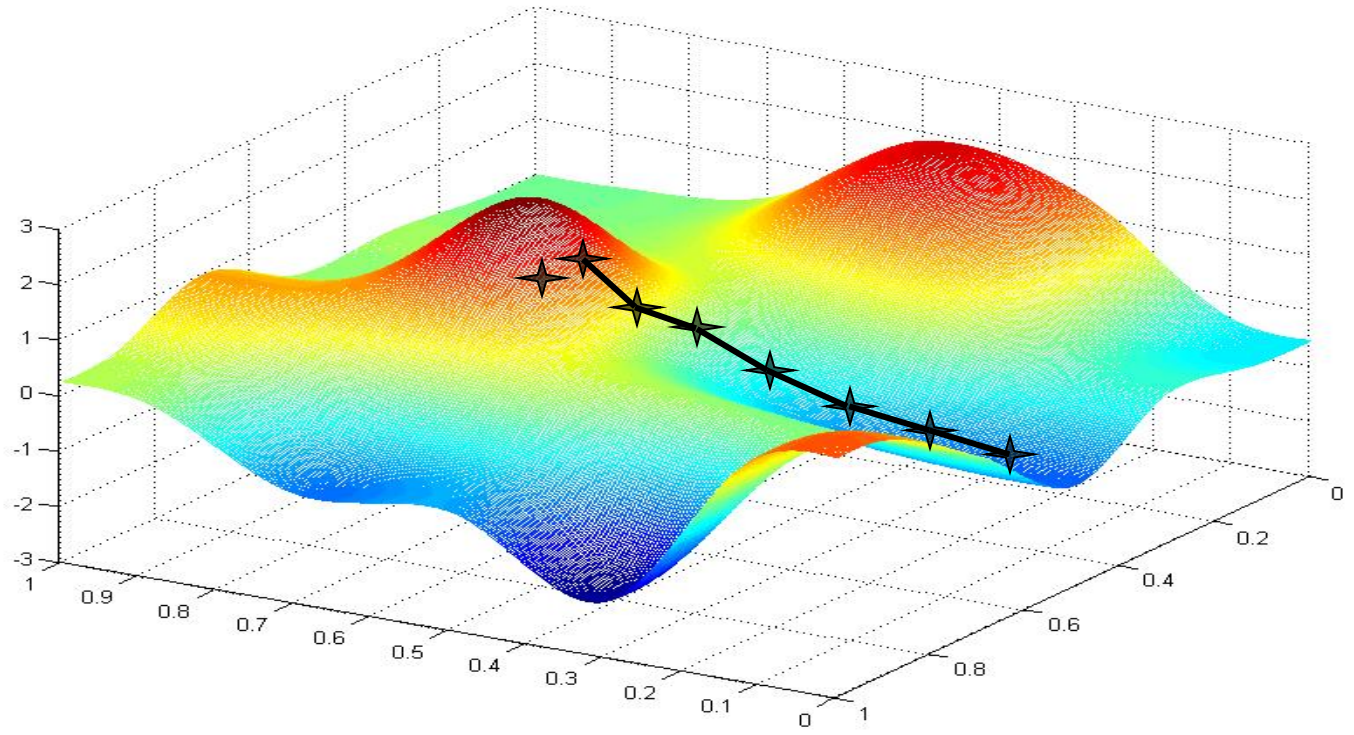
思想:

- 从某个起始点 w, b开始
- 不断改变 w, b 的值，直到找到最小值结束

单变量线性回归



单变量线性回归



单变量线性回归

- 梯度下降法

Repeat until convergence {

$$temp0 := b - \alpha \frac{\partial}{\partial b} E_{(w,b)}$$

$$temp1 := w - \alpha \frac{\partial}{\partial w} E_{(w,b)}$$

$$b := temp0$$

$$w := temp1$$

}

√

Repeat until convergence {

$$temp0 := b - \alpha \frac{\partial}{\partial b} E_{(w,b)}$$

$$b := temp0$$

$$temp1 := w - \alpha \frac{\partial}{\partial w} E_{(w,b)}$$

$$w := temp1$$

}

×

注意：要同步更新！！！

基于梯度下降法的单变量线性回归：

梯度下降法：

Repeat until convergence

{

$$temp0 := b - \alpha \frac{\partial}{\partial b} E_{(w,b)}$$

$$temp1 := w - \alpha \frac{\partial}{\partial w} E_{(w,b)}$$

$$b := temp0$$

$$w := temp1$$

}

单变量线性回归：

$$f(x_i) = w\,x_i + b$$

$$E_{(w,b)} = \frac{1}{2} \sum_{i=1}^{m} (f(x_i) - y_i)^2$$

单变量线性回归

$$\frac{\partial}{\partial w}E_{(w,b)} = \frac{\partial}{\partial w}\frac{1}{2}\sum_{i=1}^{m}(f(x_i)-y_i)^2$$

$$= \frac{\partial}{\partial w}\frac{1}{2}\sum_{i=1}^{m}(w x_i + b - y_i)^2$$

$$= \sum_{i=1}^{m}(w x_i + b - y_i)\, x_i$$

$$= \sum_{i=1}^{m}(f(x_i) - y_i)\, x_i$$

$$\frac{\partial}{\partial b}E_{(w,b)} = \frac{\partial}{\partial b}\frac{1}{2}\sum_{i=1}^{m}(f(x_i)-y_i)^2$$

$$= \sum_{i=1}^{m}\left(w x_i + b - y_i\right) \quad = \sum_{i=1}^{m}(f(x_i) - y_i)$$

单变量线性回归

基于梯度下降法的单变量线性回归：

repeat until convergence {

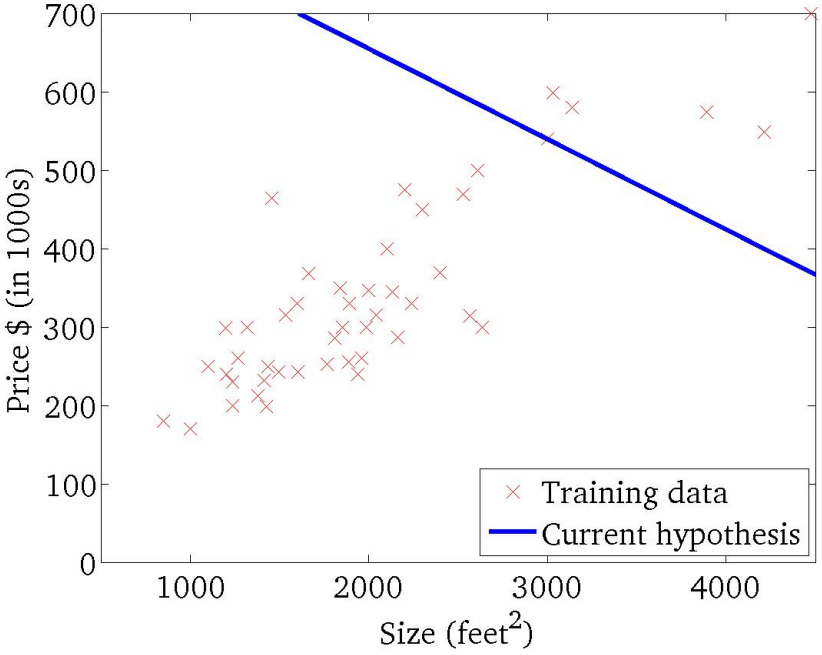
$$b := b - \alpha \sum_{i=1}^{m} (f(x_i) - y_i)$$

$$w := w - \alpha \sum_{i=1}^{m} (f(x_i) - y_i)\, x_i$$

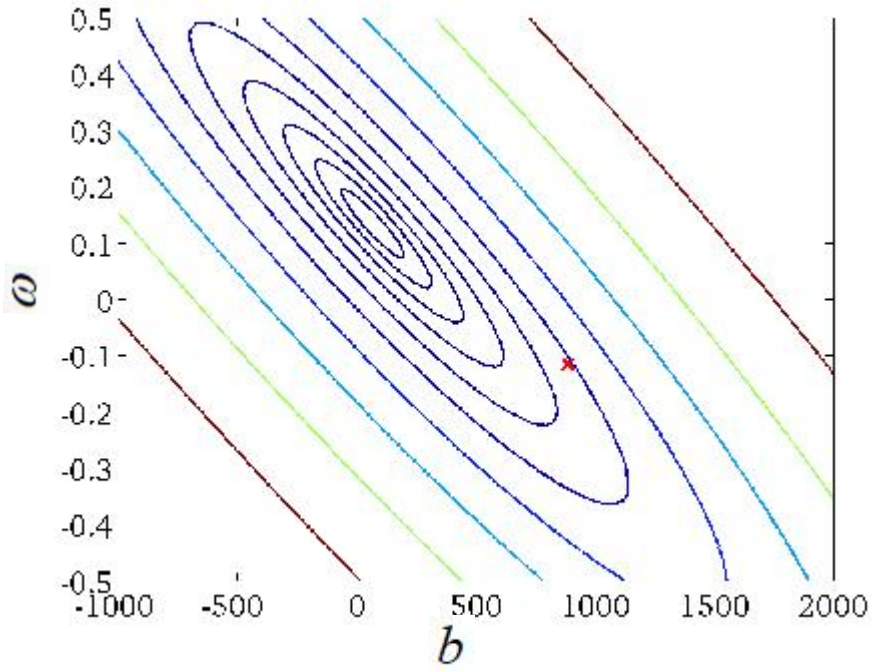
} 注意：同时更新b和w!!!

批量梯度下降：每一步都使用了所有训练样本

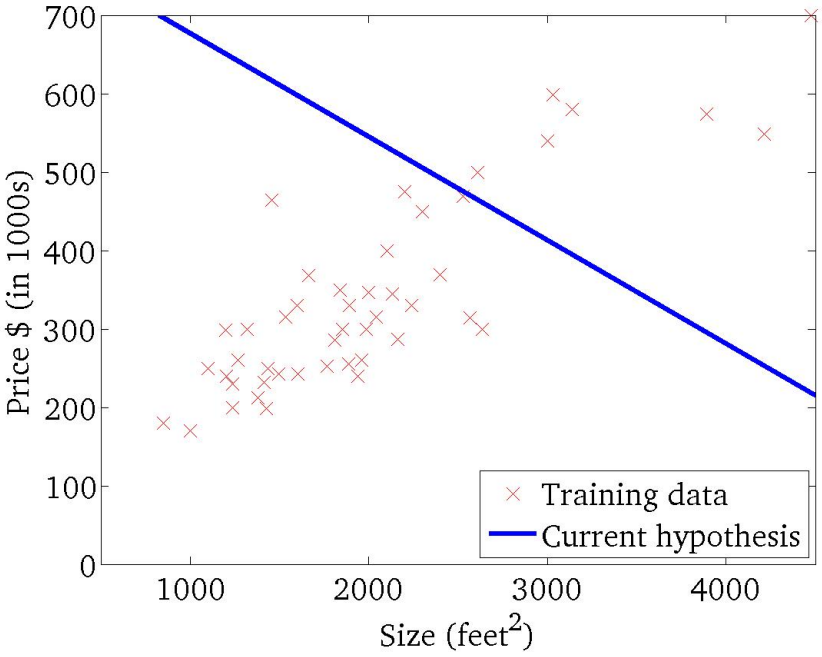
$f(x)$


Price $ (in 1000s)
Size (feet^2)
Training data
Current hypothesis
0
100
200
300
400
500
600
700
1000
2000
3000
4000

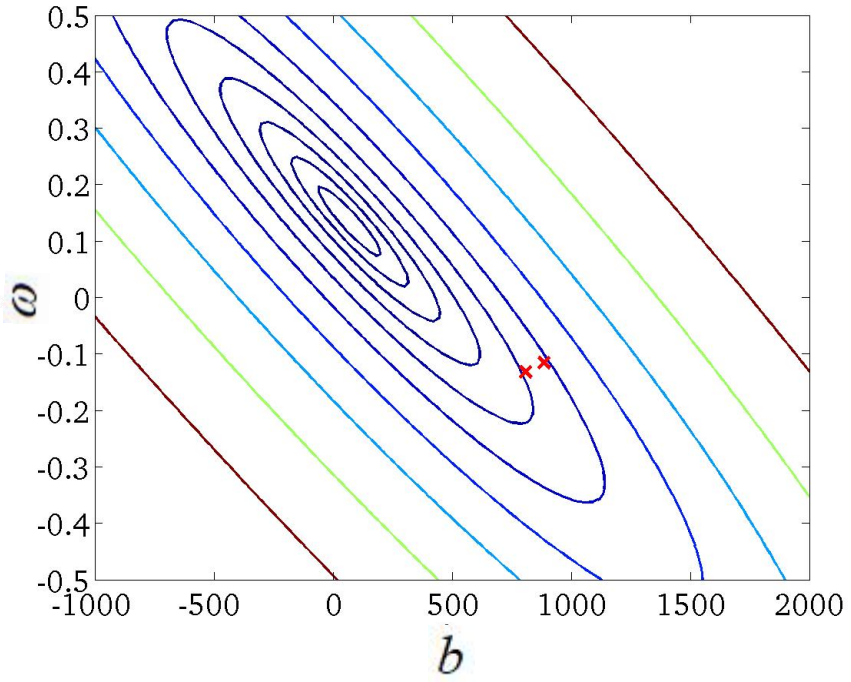
$E_{(w,b)}$


w
b
0.5
0.4
0.3
0.2
0.1
0
-0.1
-0.2
-0.3
-0.4
-0.5
-1000
-500
0
500
1000
1500
2000

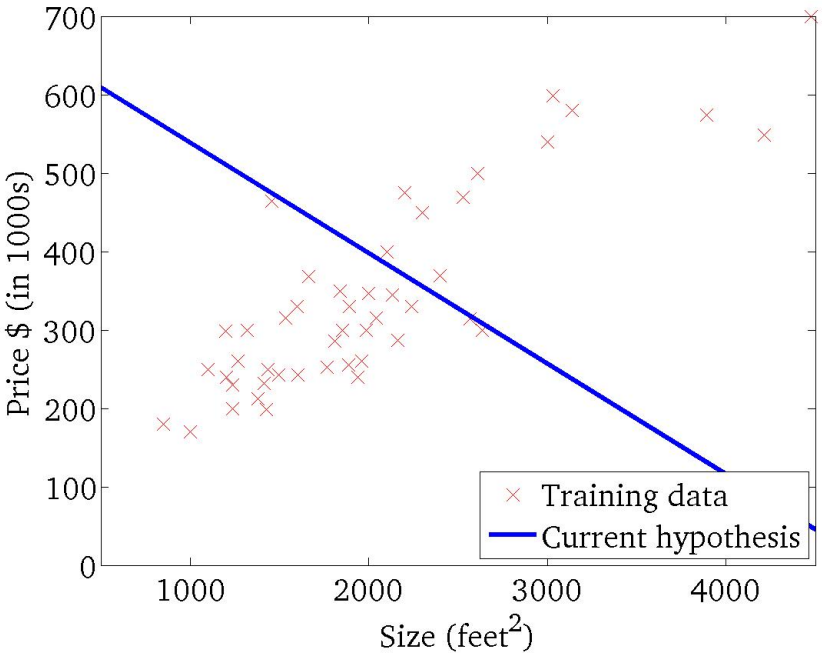
$f(x)$


Price $ (in 1000s)
Size (feet^2)
Training data
Current hypothesis
0
100
200
300
400
500
600
700
1000
2000
3000
4000

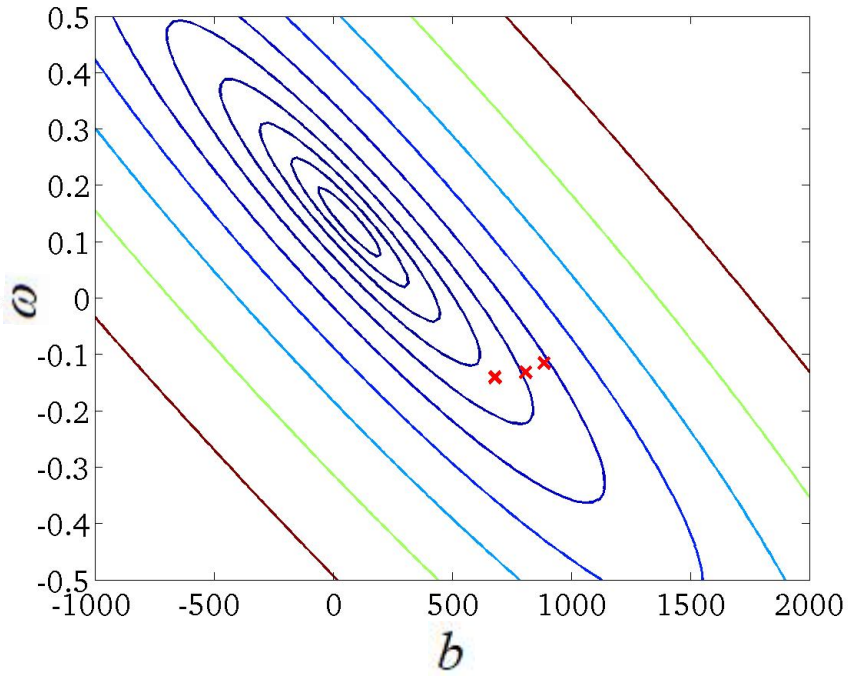
$E_{(w,b)}$


w
b
0.5
0.4
0.3
0.2
0.1
0
-0.1
-0.2
-0.3
-0.4
-0.5
-1000
-500
0
500
1000
1500
2000

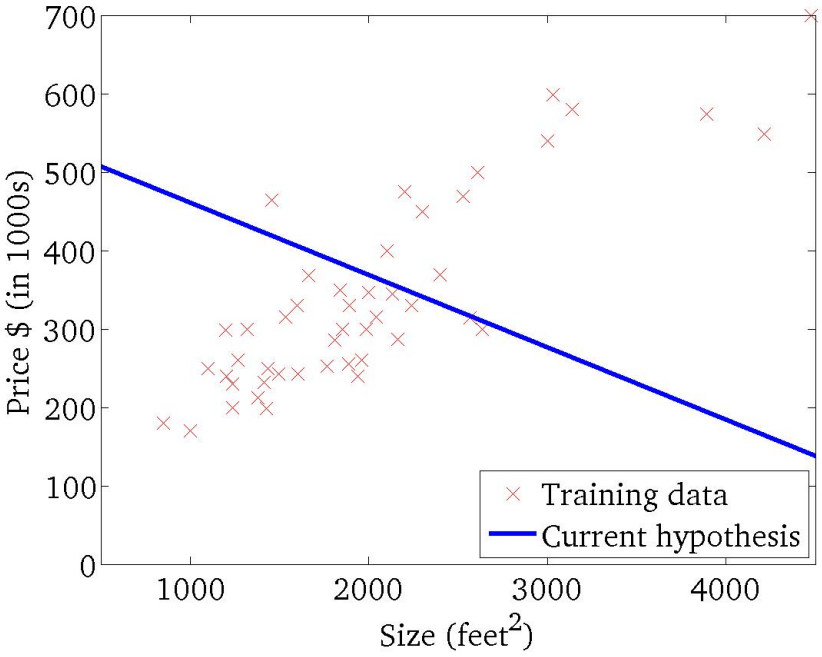
$f(x)$


Price $ (in 1000s)
Size (feet^2)
Training data
Current hypothesis

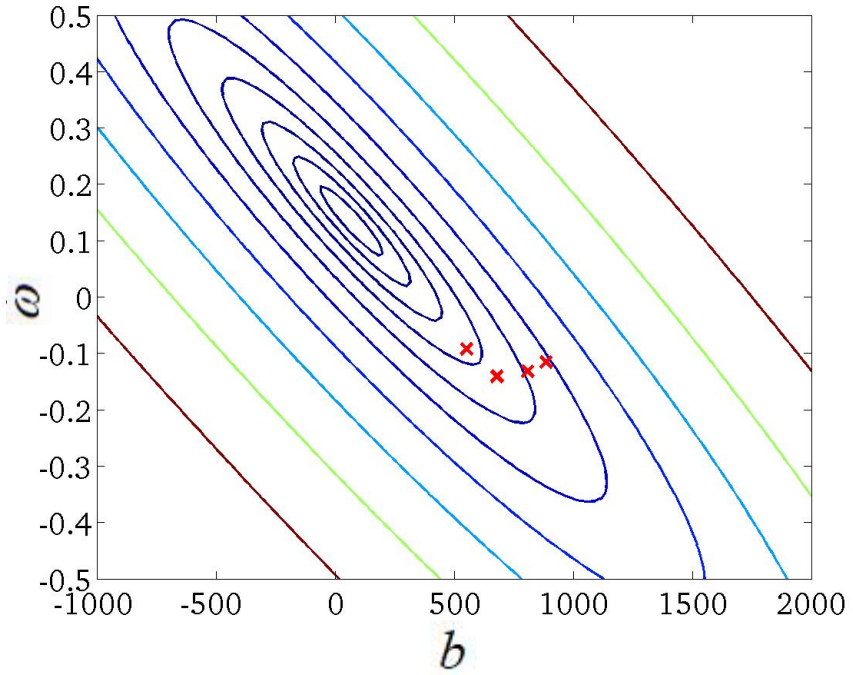
$E_{(w,b)}$


w
b

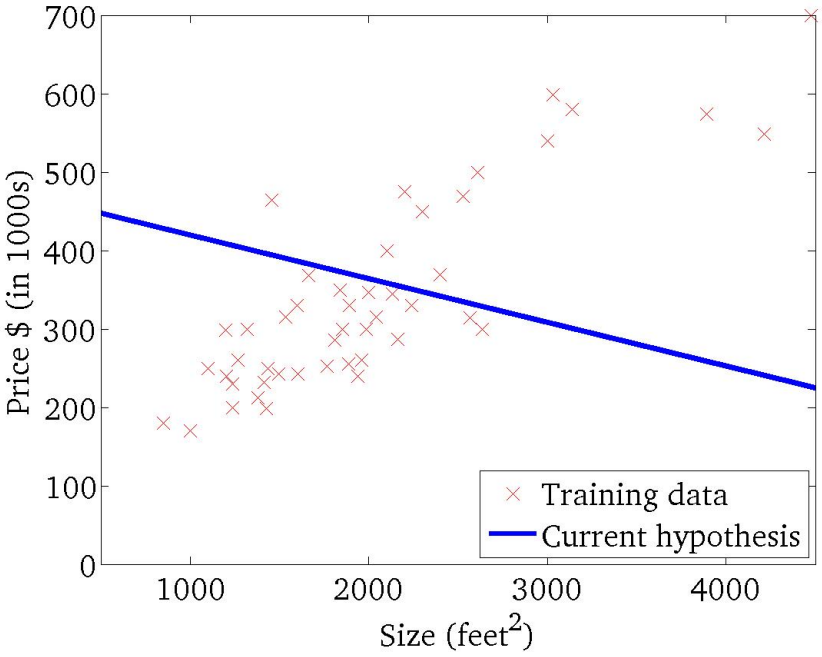
$f(x)$


Price $ (in 1000s)
Size (feet^2)
Training data
Current hypothesis
0
100
200
300
400
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3000
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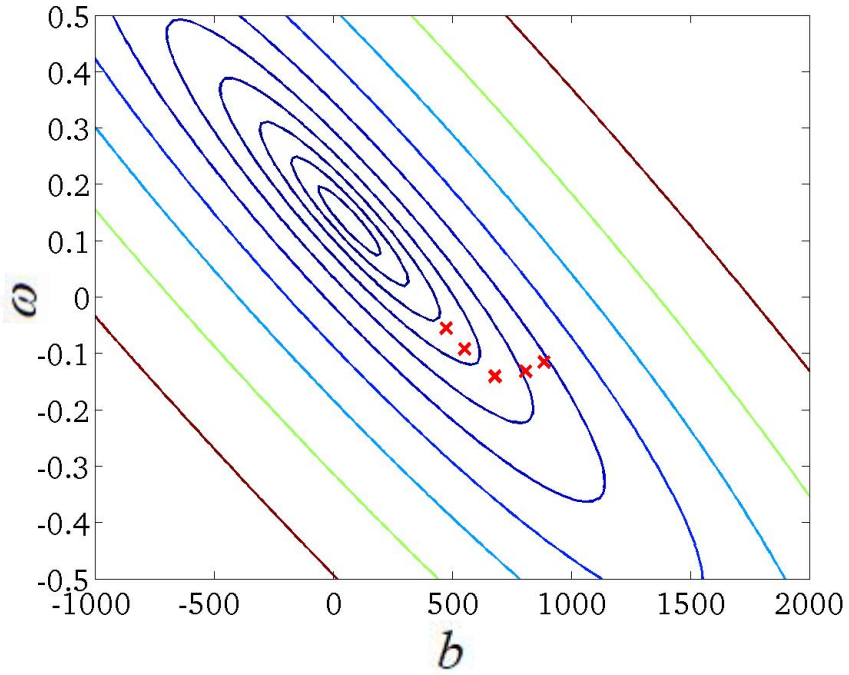
$E_{(w,b)}$


w
b
0.5
0.4
0.3
0.2
0.1
0
-0.1
-0.2
-0.3
-0.4
-0.5
-1000
-500
0
500
1000
1500
2000

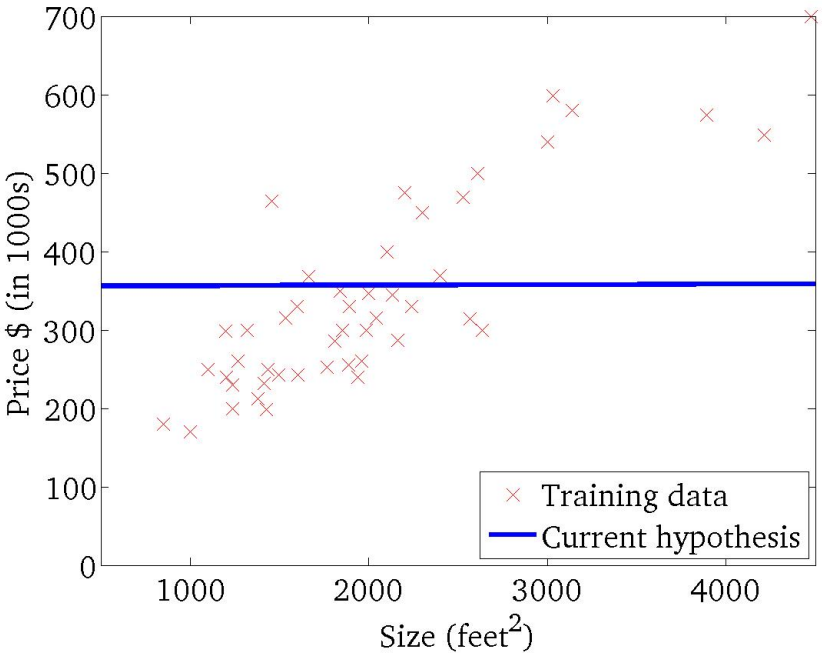
$f(x)$


Price $ (in 1000s)
Size (feet^2)
Training data
Current hypothesis

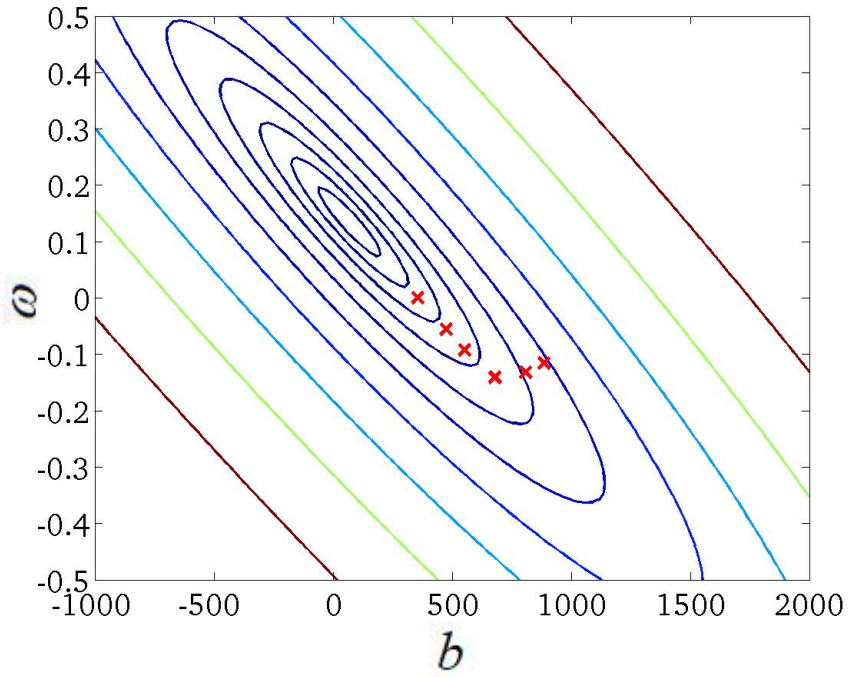
$E_{(w,b)}$


w
b

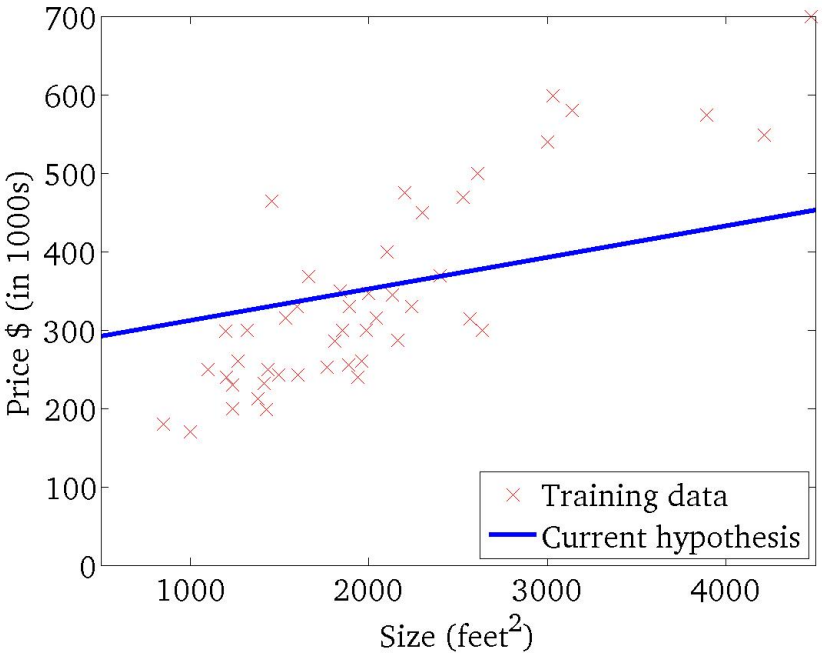
$f(x)$


Price $ (in 1000s)
Size (feet^2)
Training data
Current hypothesis
0
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200
300
400
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2000
3000
4000

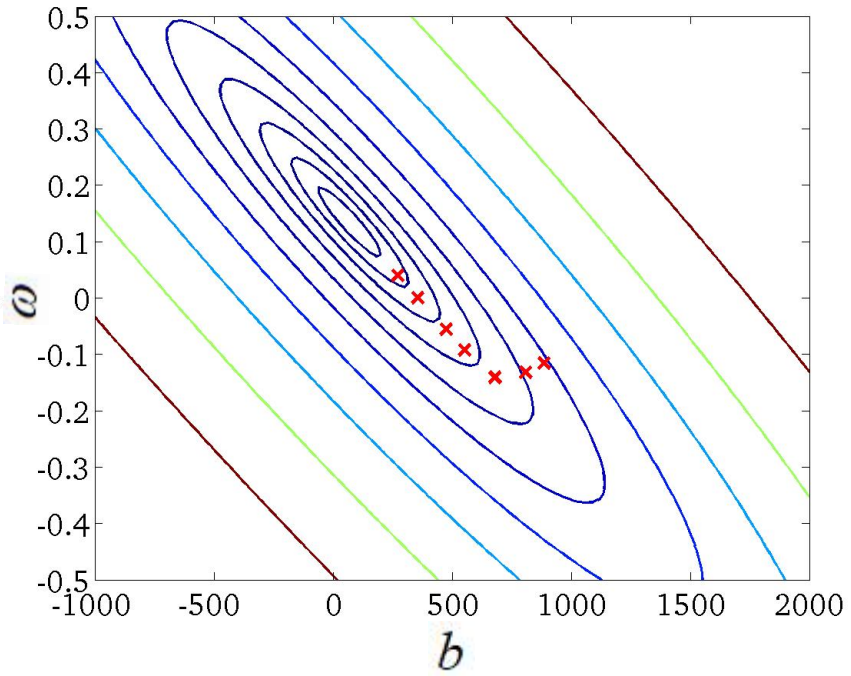
$E_{(w,b)}$


w
b
0.5
0.4
0.3
0.2
0.1
0
-0.1
-0.2
-0.3
-0.4
-0.5
-1000
-500
0
500
1000
1500
2000

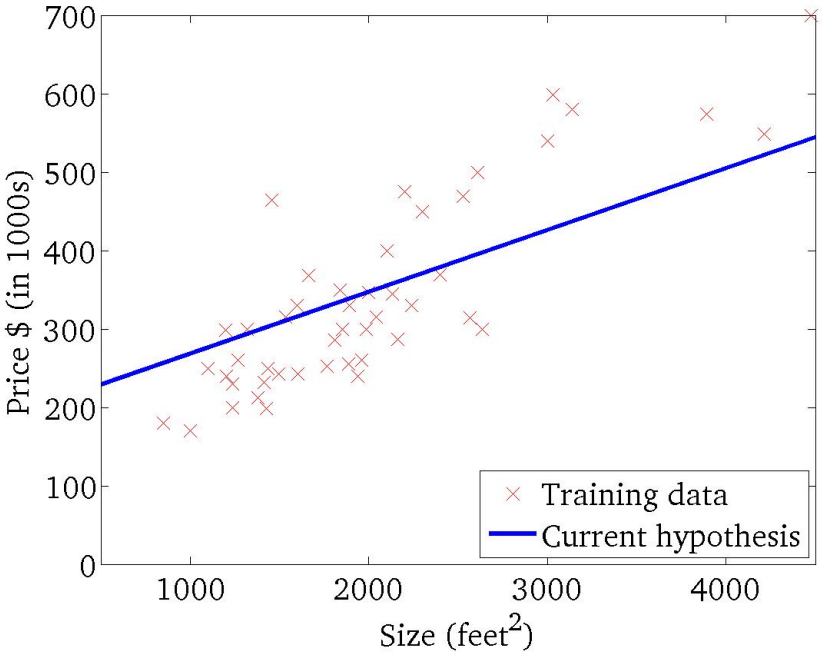
$f(x)$


Price $ (in 1000s)
Size (feet^2)
Training data
Current hypothesis
0
100
200
300
400
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3000
4000

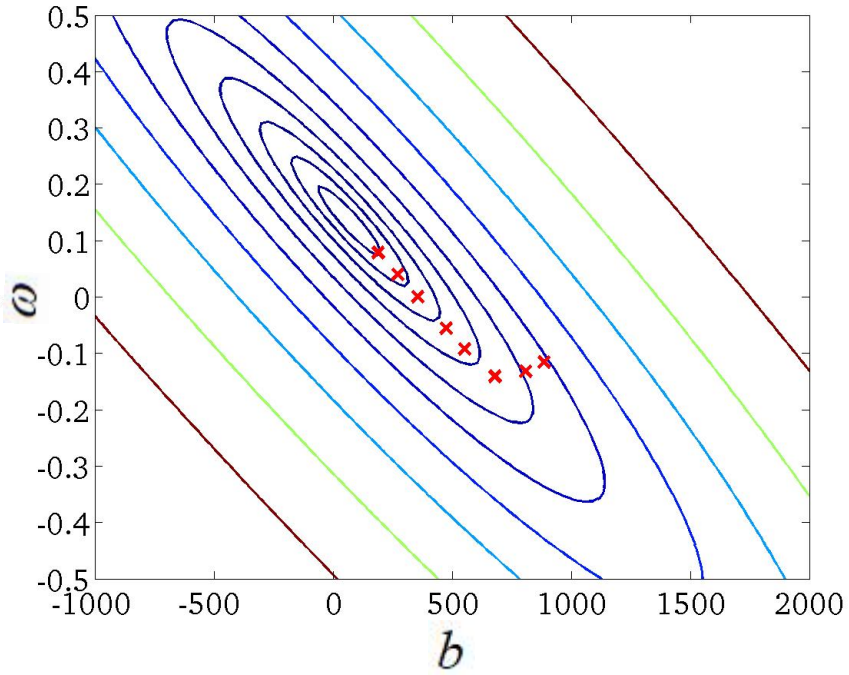
$E_{(w,b)}$


w
b
0.5
0.4
0.3
0.2
0.1
0
-0.1
-0.2
-0.3
-0.4
-0.5
-1000
-500
0
500
1000
1500
2000

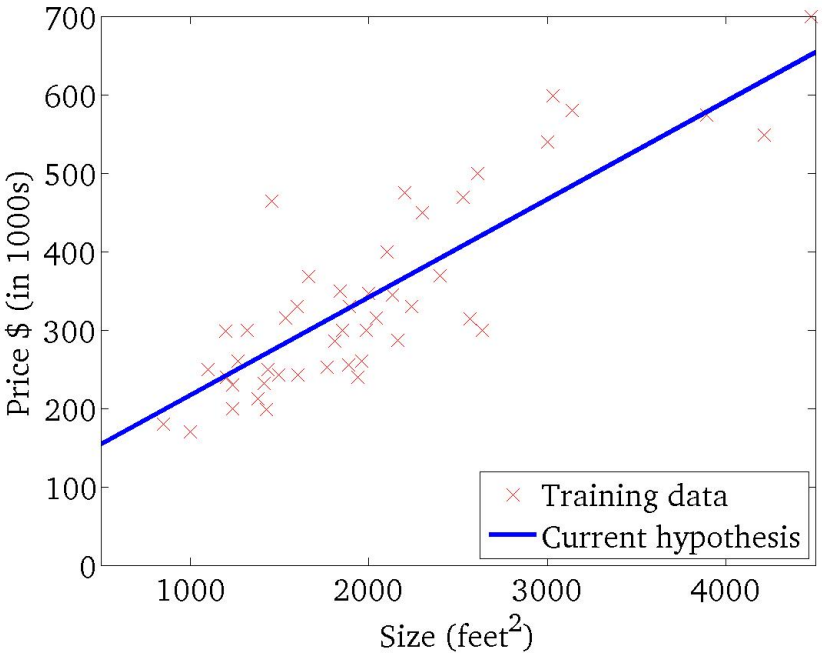
$f(x)$


Price $ (in 1000s)
Size (feet^2)
Training data
Current hypothesis
0
100
200
300
400
500
600
700
1000
2000
3000
4000

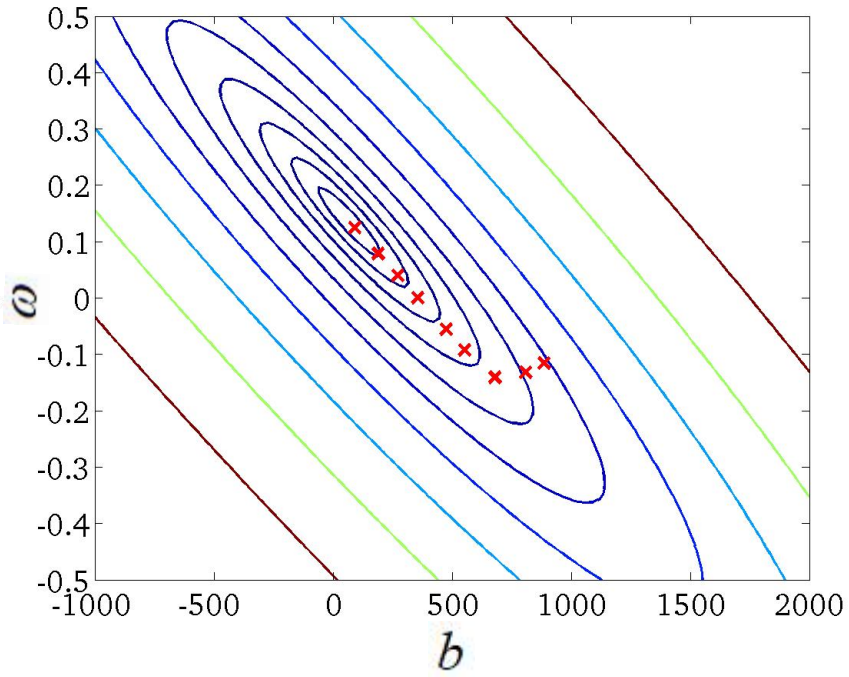
$E_{(w,b)}$


w
b
0.5
0.4
0.3
0.2
0.1
0
-0.1
-0.2
-0.3
-0.4
-0.5
-1000
-500
0
500
1000
1500
2000

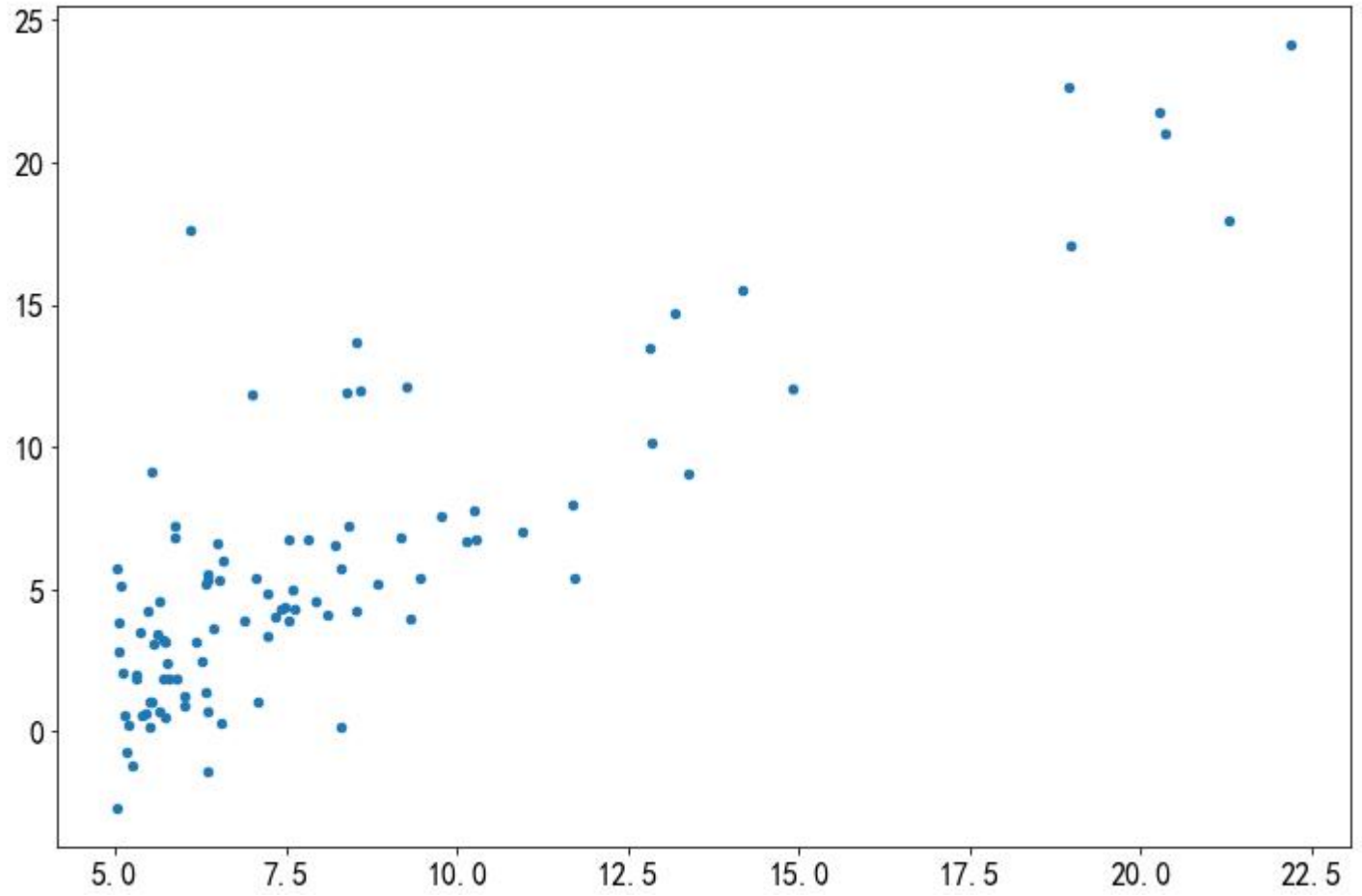
$f(x)$

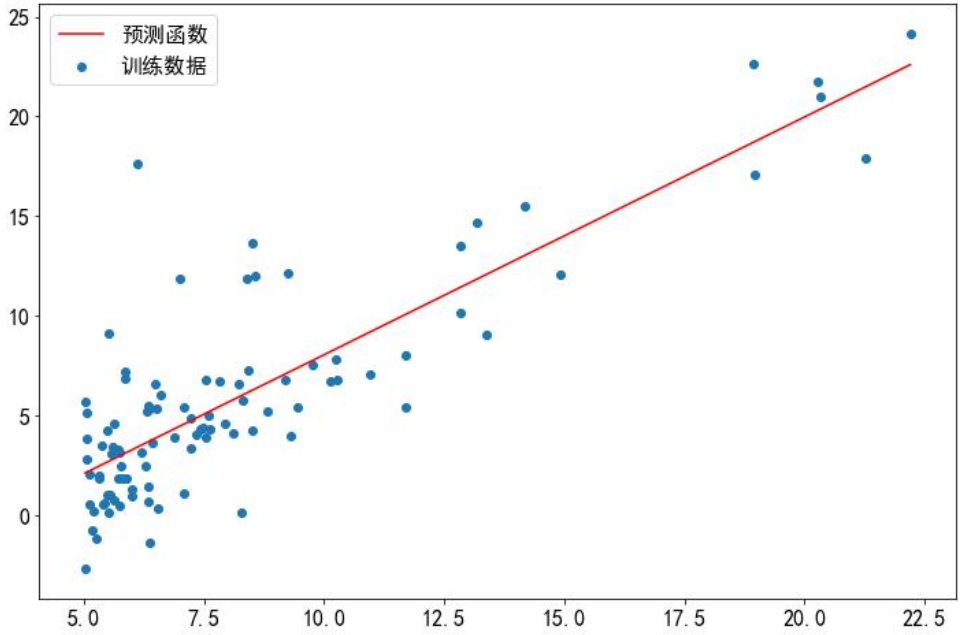

Price $ (in 1000s)
Size (feet^2)
Training data
Current hypothesis
0
100
200
300
400
500
600
700
1000
2000
3000
4000

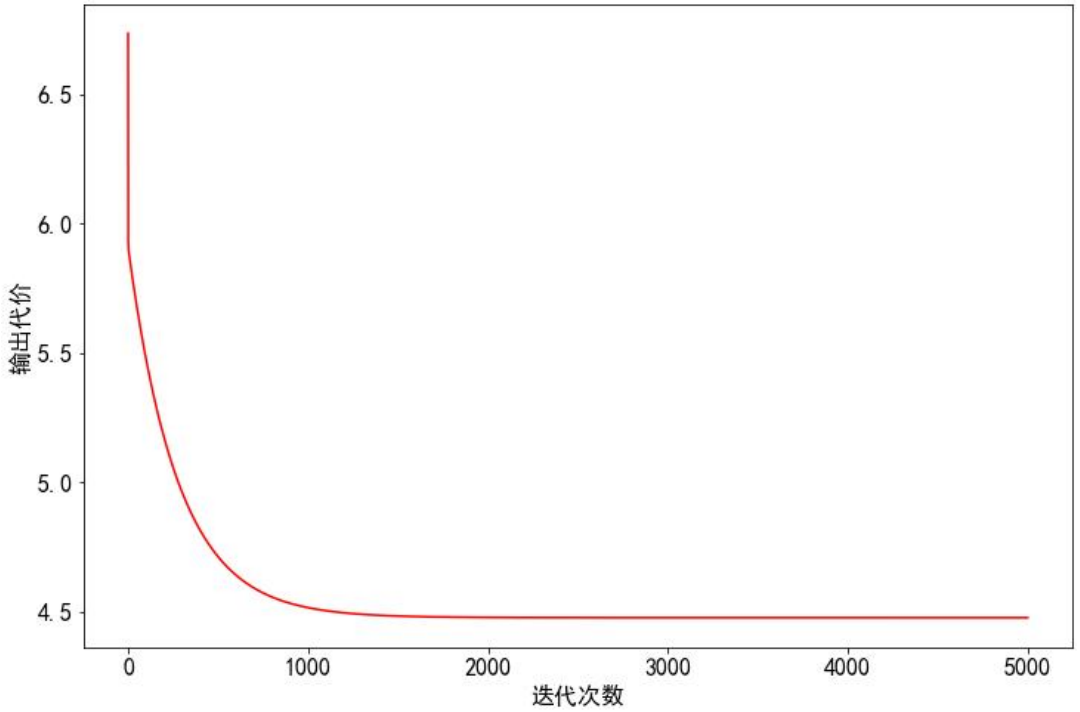
$E_{(w,b)}$


w
b
0.5
0.4
0.3
0.2
0.1
0
-0.1
-0.2
-0.3
-0.4
-0.5
-1000
-500
0
500
1000
1500
2000

示例




预测函数
训练数据
25
20
15
10
5
0
5.0
7.5
10.0
12.5
15.0
17.5
20.0
22.5


6.5
6.0
5.5
5.0
4.5
输出代价
0
1000
2000
3000
4000
5000
迭代次数

多变量线性回归

$$y_i = f(\boldsymbol{x}_i) = w_1 x_{i1} + w_2 x_{i2} + \cdots + w_n x_{in} + b = \boldsymbol{w}^T \boldsymbol{x}_i + b$$

其中：
$$\boldsymbol{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} \qquad \boldsymbol{x}_i = \begin{bmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{in} \end{bmatrix}$$

或记作：
$$y_i = f(\boldsymbol{x}_i) = w_1 x_{i1} + w_2 x_{i2} + \cdots + w_n x_{in} + b$$

令$w_0 = b$，$x_{i0} = 1$

$$= w_0 x_{i0} + w_1 x_{i1} + \cdots + w_n x_{in}$$

$$= \mathrm{w}^{\mathrm{T}} \mathrm{x}_{\mathrm{i}}$$

其中：
$$\mathrm{w} = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_n \end{bmatrix} \qquad \mathrm{x}_i = \begin{bmatrix} x_{i0} \\ x_{i1} \\ \vdots \\ x_{in} \end{bmatrix} \qquad \omega_0 = b \qquad x_{i0} = 1$$

多变量线性回归

Size (feet²) x_1	Number of bedrooms x_2	Number of floors x_3	Age of home (years) x_4	Price ($1000) y
2104	5	1	45	460
1416	3	2	40	232
1534	3	2	30	315
852	2	1	36	178
...	...	...	...	...

n：#特征

m：#训练样本

$\boldsymbol{x}_i$ ： 第i个训练样本

x_{ij} ： 第i个训练样本的第j个特征值

$$\boldsymbol{x}_2 = \begin{bmatrix} x_{21} \\ x_{22} \\ x_{23} \\ x_{24} \end{bmatrix} = \begin{bmatrix} 1416 \\ 3 \\ 2 \\ 40 \end{bmatrix}$$

多变量线性回归

$n = 4 \quad m = 4$

Size (feet²) x_1	Number of bedrooms x_2	Number of floors x_3	Age of home (years) x_4	Price ($1000) y
2104	5	1	45	460
1416	3	2	40	232
1534	3	2	30	315
852	2	1	36	178

多变量线性回归

$n = 4 \quad m = 4$

	Size (feet²)	Number of bedrooms	Number of floors	Age of home (years)	Price ($1000)
x_0	x_1	x_2	x_3	x_4	y
1	2104	5	1	45	460
1	1416	3	2	40	232
1	1534	3	2	30	315
1	852	2	1	36	178

$$\mathbf{X} = \begin{bmatrix} 1 & 2104 & 5 & 1 & 45 \\ 1 & 1416 & 3 & 2 & 40 \\ 1 & 1534 & 3 & 2 & 30 \\ 1 & 852 & 2 & 1 & 36 \end{bmatrix}_{m\times(n+1)} = \begin{bmatrix} 1 & \boldsymbol{x}_1^{\,T} \\ 1 & \boldsymbol{x}_2^{\,T} \\ \vdots & \vdots \\ 1 & \boldsymbol{x}_m^{\,T} \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} 460 \\ 232 \\ 315 \\ 178 \end{bmatrix}_{m\times 1} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix} \quad \mathbf{w} = \begin{bmatrix} b \\ w_1 \\ w_2 \\ w_3 \\ w_4 \end{bmatrix}_{(n+1)\times 1} = \begin{bmatrix} b \\ w_1 \\ \vdots \\ w_n \end{bmatrix}$$

$$\mathbf{y} \approx f(\mathbf{X}) = \mathbf{X}\mathbf{w} \qquad E_{(\mathbf{w})} = \frac{1}{2}\sum_{i=1}^{m}(f(\boldsymbol{x}_i) - y_i)^2$$

多变量线性回归

$$E_{(\mathbf{w})} = \frac{1}{2}\sum_{i=1}^{m}(f(\boldsymbol{x}_i)-y_i)^2 = \frac{1}{2}\sum_{i=1}^{m}(\mathbf{w}^T\mathbf{x}_i - y_i)^2 = \frac{1}{2}\sum_{i=1}^{m}(\mathbf{x}_i{}^T\mathbf{w} - y_i)^2$$

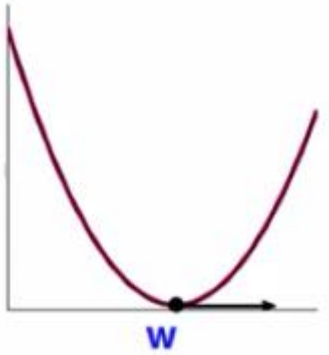
$$= \frac{1}{2}\left\| \begin{bmatrix} \mathbf{x}_1{}^T\mathbf{w} - y_1 \\ \mathbf{x}_2{}^T\mathbf{w} - y_2 \\ \vdots \\ \mathbf{x}_m{}^T\mathbf{w} - y_m \end{bmatrix} \right\|^2 = \frac{1}{2}\left\| \underbrace{\begin{bmatrix} \mathbf{x}_1{}^T \\ \mathbf{x}_2{}^T \\ \vdots \\ \mathbf{x}_m{}^T \end{bmatrix}}_{\mathbf{X}}\mathbf{w} - \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}}_{\mathbf{y}} \right\|^2$$

$$= \frac{1}{2}\left\| X_{m\times(n+1)}\mathbf{w}_{(n+1)\times 1} - \mathbf{y}_{m\times 1} \right\|^2$$

多变量线性回归

$$E_{(\mathbf{w})} = \frac{1}{2}\left\|\mathbf{Xw}-\mathbf{y}\right\|^2$$

连续，可微，凸函数



$$\nabla E_{(\mathbf{w})} = \begin{bmatrix} \frac{\partial}{\partial w_0} E_{(\mathbf{w})} \\ \frac{\partial}{\partial w_1} E_{(\mathbf{w})} \\ \vdots \\ \frac{\partial}{\partial w_n} E_{(\mathbf{w})} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$E_{(\mathbf{w})} = \frac{1}{2}\left\|\mathbf{Xw}-\mathbf{y}\right\|^2 = \frac{1}{2}\left(\mathbf{Xw}-\mathbf{y}\right)^T\left(\mathbf{Xw}-\mathbf{y}\right)$$

$$= \frac{1}{2}\left(\mathbf{w}^T\mathbf{X}^T\mathbf{Xw}-\mathbf{w}^T\mathbf{X}^T\mathbf{y}-\mathbf{y}^T\mathbf{Xw}+\mathbf{y}^T\mathbf{y}\right)$$

$$\nabla E_{(\boldsymbol{w})} = \nabla\left(\frac{1}{2}\left(\mathbf{w}^T\mathbf{X}^T\mathbf{X}\mathbf{w} - \mathbf{w}^T\mathbf{X}^T\boldsymbol{y} - \boldsymbol{y}^T\mathbf{X}\mathbf{w} + \boldsymbol{y}^T\boldsymbol{y}\right)\right)$$

$$= \frac{1}{2}\left(2\mathbf{X}^T\mathbf{X}\mathbf{w} - \mathbf{X}^T\boldsymbol{y} - \left(\boldsymbol{y}^T\mathbf{X}\right)^T\right) = \mathbf{X}^T\mathbf{X}\mathbf{w} - \mathbf{X}^T\boldsymbol{y}$$

$$= \mathbf{0}$$

$$\Rightarrow \quad \mathbf{w} = \underbrace{\left(\mathbf{X}^T\mathbf{X}\right)^{-1}}_{\mathbf{X}^+}\mathbf{X}^T\boldsymbol{y} = \mathbf{X}^+\mathbf{y}$$

多变量线性回归

假设： $y_i = f(\boldsymbol{x}_i) = w_1 x_{i1} + w_2 x_{i2} + \cdots + w_n x_{in} + b = \mathbf{w}^T \mathbf{x}_i$

参数： $w_0, w_1, \quad w_2, \cdots, \quad w_n$

代价函数： $E_{(\boldsymbol{w})} = E_{(w_0, w_1, \cdots w_n)} = \frac{1}{2} \sum_{i=1}^{m} (f(\boldsymbol{x}_i) - y_i)^2$

***n*=1：**

重复直到收敛 {

$$w_j := w_j - \alpha \sum_{i=1}^{m} (f(x_i) - y_i)\ x_i$$

(同时更新$w_j, \quad j = 0,1$)

$= \frac{\partial E_{(\boldsymbol{w})}}{\partial w_j}$

}

***n*>1：**

重复直到收敛 {

$= \frac{\partial E_{(\boldsymbol{w})}}{\partial w_j}$

$$w_j := w_j - \alpha \sum_{i=1}^{m} (f(\boldsymbol{x}_i) - y_i)\ x_{ij}$$

(同时更新$w_j, \quad j = 0,1, \cdots, \quad n$)

}